

INT217: INTRODUCTION TO DATA MANAGEMENT

(Project Semester January-April 2025)

(TITLE OF THE PROJECT)

**Consumer Complain Analysis Dashboard: Analyzing Complain Status,
Complaints by Company, and Complain by products...all these things**

Submitted by

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Section – K23GW

Programme - B.tech CSE

Course Code- INT 217

Under the Guidance of

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Discipline of CSE

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CERTIFICATE

This is to certify that Shezal bearing Registration no. 12319819 has completed INT 217 project titled, “Consumer Complaints Analysis Dashboard” - To the best of my knowledge, the present work is the result of his original development, effort, and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 13-04-2024

DECLARATION

I, Ms. Shezal, student of Computer Science and Engineering under CSE Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 13-04-2025

Registration No. - **12319819**

Name of the student - **Shezal**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, Ms. Baljinder Kaur, for her valuable guidance, support, and encouragement throughout the completion of this project.

I also thank my friends for their continuous motivation and helpful discussions during this journey.

Lastly, I acknowledge my own dedication and consistent efforts that helped me bring this project to completion.

Shezal

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1. Introduction

This project is based on data from the **Consumer Complaint Database** provided by the U.S. government. The purpose of this project is to analyze complaints submitted by consumers about different financial products and services. These complaints help us understand what problems people are facing with banks, credit cards, loans, and other financial activities.

In this project, I used Microsoft Excel to clean the data and create pivot tables and an interactive dashboard. The dashboard helps to easily understand the data by showing visuals like charts, graphs, and maps. It covers key areas such as:

- Complaints by product (like credit cards or loans)
- Complaints by issue (like billing problems or wrong charges)
- Complaints by company

- How complaints were submitted (web, phone, etc.)
- Whether the company gave a timely response or not
- How complaints were resolved
- State-wise distribution of complaints

The goal of this project is to find which companies and products get the most complaints, what the most common issues are, and how complaints are being handled. This information can help companies improve their services and help people become more aware of financial problems.

2. Source of Dataset

The dataset used for this project was downloaded from the official U.S. government open data website, data.gov. It comes from the **Consumer Complaint Database**, which is managed by the Consumer Financial Protection Bureau (CFPB). This database contains real complaints submitted by consumers about financial products and services.

Dataset Link:

<https://catalog.data.gov/dataset/consumer-complaint-database>

Key Features of the Dataset:

- **Product Type:** Complaints related to credit cards, checking accounts, loans, virtual currencies, etc.
- **Issue:** Describes the specific problem (e.g., incorrect charges, fraud, loan denial).
- **Company Name:** The financial institution the complaint is about.
- **Submitted Via:** How the complaint was sent (web, phone, referral, or mail).
- **Timely Response:** Whether the company responded on time or not.
- **Complaint Resolution:** Whether the complaint was closed with or without explanation, or if any relief was provided.
- **State:** The U.S. state from which the complaint was submitted.

The dataset is large, regularly updated, and well-organized, making it ideal for analysis. I used Excel to clean the data, organize it using pivot tables, and create a dashboard that displays the results in charts and graphs for easier understanding.

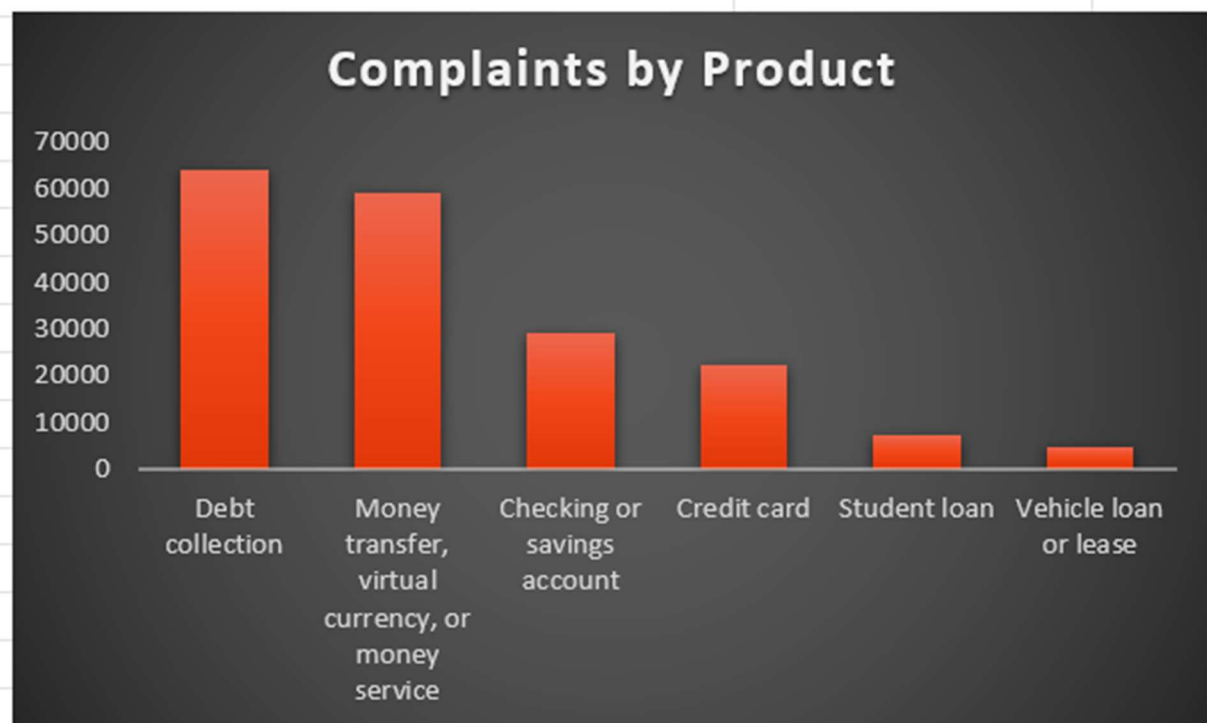
3. Pivot Table Analysis

In this project, I used several pivot tables to analyze the complaints data from different angles. These tables helped me break down and understand the data more clearly. Here's how each one helped:

- **Complaints by Product:** Shows which products had the most complaints.
- **Complaints by States:** Highlights which states received the highest number of complaints.
- **Complaints by Issue:** Tells us the most common problems customers faced.
- **Submitted via / Complaints via:** Shows how people submitted their complaints (online, phone, etc.).
- **Complaints Status:** Displays the current status of complaints (e.g., closed, in progress).
- **Complaints by Company:** Identifies which companies received the most complaints.

These pivot tables allowed me to create a clear and interactive dashboard where users can quickly understand the data and gain insights with just a few clicks.

2		
3	Row Labels	Count of Complaint ID
4	Debt collection	64288
5	Money transfer, virtual currency, or money service	58968
6	Checking or savings account	29073
7	Credit card	22374
8	Student loan	7367
9	Vehicle loan or lease	4651



Row Labels	Count of Complaint ID
Problem with a purchase shown on your statement	4114
Dealing with your lender or servicer	4307
Problem with a company's investigation into an existing problem	4357
Took or threatened to take negative or legal action	6602
Problem caused by your funds being low	8756
False statements or representation	9275
Managing an account	13934
Written notification about debt	14600
Attempts to collect debt not owed	29695
Other transaction problem	47469



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Row Labels		Count of Complaint ID		
Web	181612		Web	181612
Phone	2455		Phone	2455
Referral	2129		Referral	2129
Postal mail	525		Postal mai	525



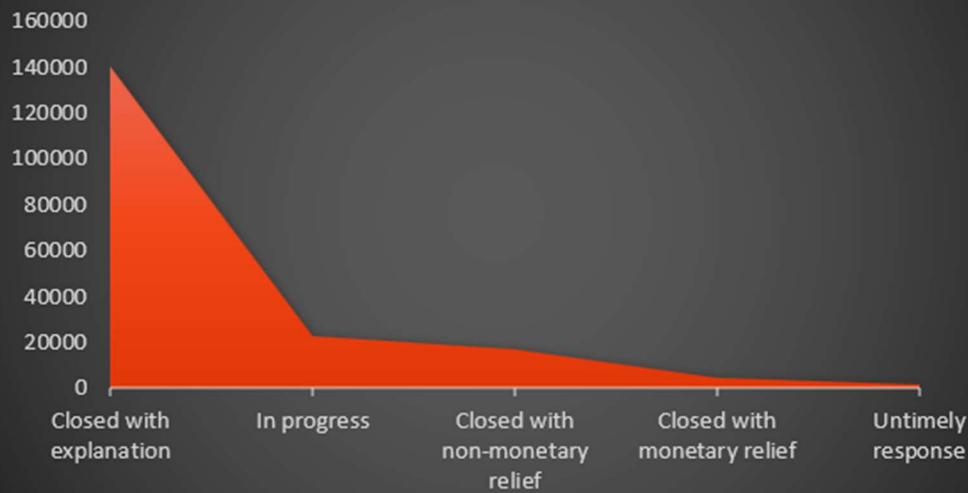
Row Labels

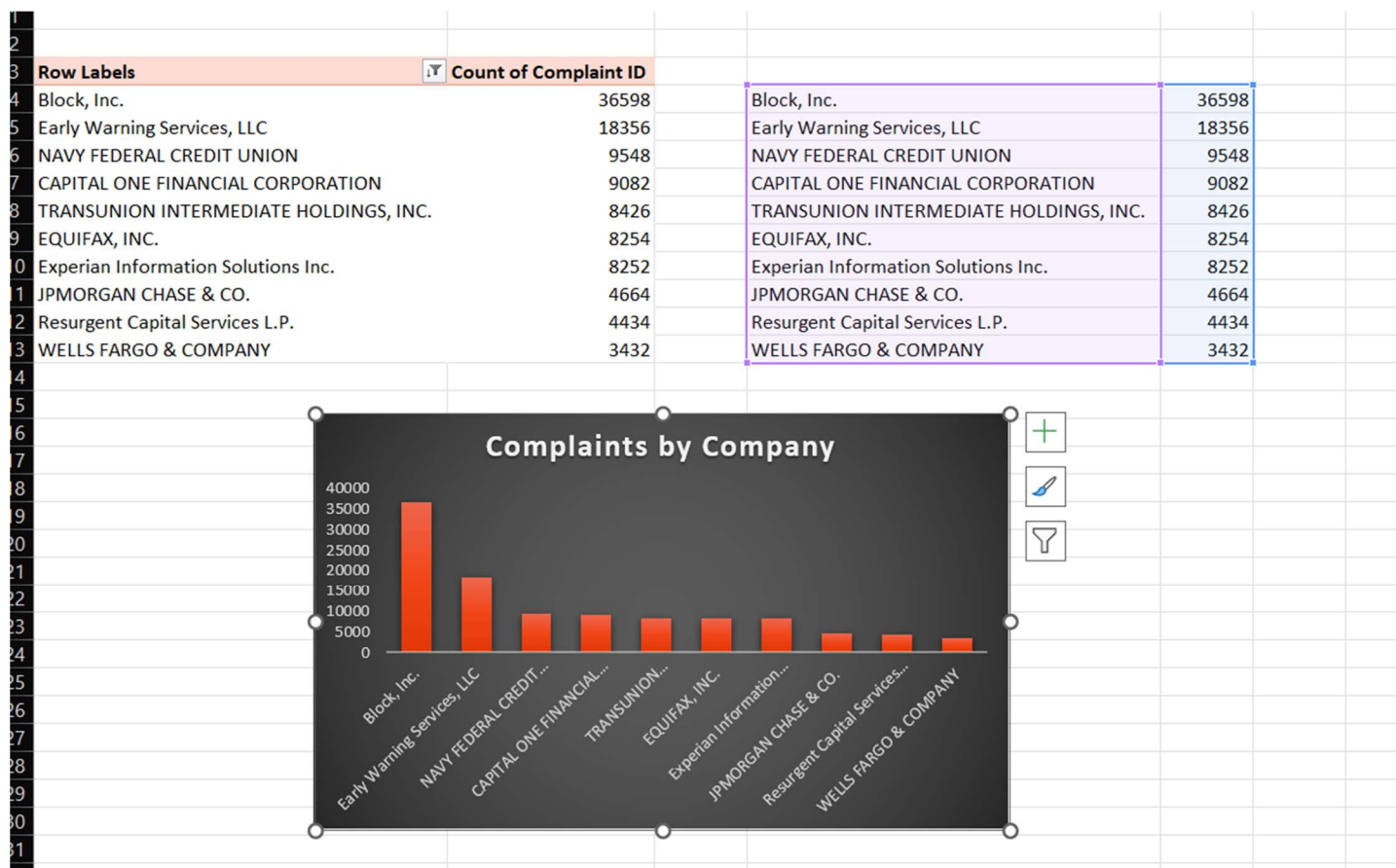
Count of Complaint ID

Closed with explanation	140809
In progress	22662
Closed with non-monetary relief	17307
Closed with monetary relief	4604
Untimely response	1339

Closed with explanation	140809
In progress	22662
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Complaints Status





4. Analysis on Dataset

i. Introduction

The main goal of this analysis is to understand trends in consumer complaints across the United States. This includes identifying which products receive the most complaints, which companies are involved, and how complaints are being resolved. An Excel dashboard was created to display these insights in a clear and interactive way.

ii. General Description

The dataset contains hundreds of thousands of records from consumers who submitted complaints about financial products and services. The data includes details such as product type, complaint issue, company name, method of complaint submission, and how the company responded. The complaints span across different years and states.

iii. Analysis Techniques and Dashboard Features

- **Pivot Tables:** Used to summarize complaints by product, company, state, and response type.
- **Slicers:** product type, company, and state, and complaints/submitted via...

- **Interactive Dashboard:** Includes charts, slicers, and visual elements that update based on user selections.

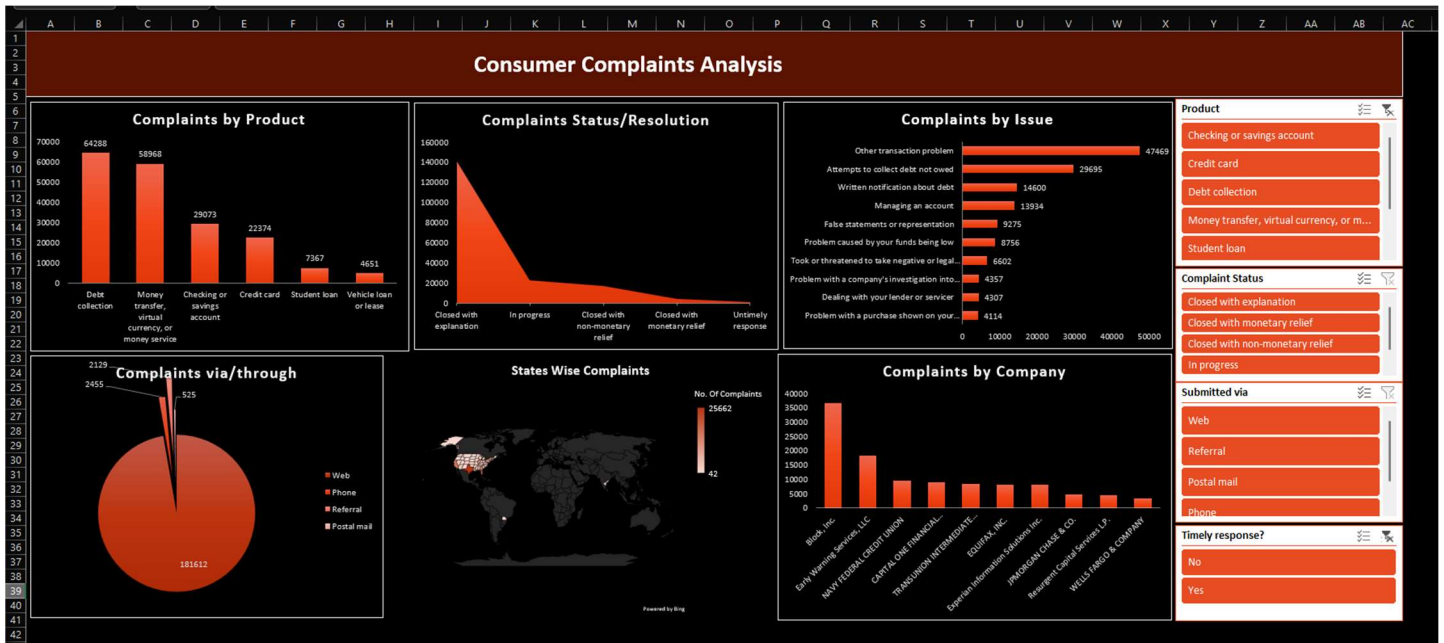
iv. Key Findings and Results

- **Most Complained Products:** Credit reporting, credit card, and mortgage services received the highest number of complaints.
- **Top Companies:** Some large financial institutions appeared frequently in the dataset, indicating repeated consumer issues.
- **Submission Method:** Most complaints were submitted through the website, showing consumer preference for online channels.
- **Response Timeliness:** A large number of companies responded on time, but some still failed to provide a timely reply.
- **Complaint Outcomes:** Many complaints were closed with an explanation, while others were closed without relief or action.

v. Visualizations

- **Bar Charts:** Show the top 5 most complained-about products and companies.
- **Pie Charts:** Display the percentage of complaints by submission method or complaint status.
- **Line Graphs:** Track complaint trends over months or years.
- **Stacked Bars:** Compare how different companies responded to complaints across multiple categories.

(Note: Actual charts would be included in the Excel dashboard. Descriptions are based on typical visualizations for this data.)



5. Conclusion

The analysis of consumer complaints using pivot tables and an interactive Excel dashboard has helped uncover important patterns in how people experience and report problems with financial products and services. By looking at the data from different angles—such as by product type, company, and complaint outcome—we gained useful insights into common issues faced by consumers.

Key takeaways include:

- **Credit reporting and credit card complaints** are among the most common, showing the need for improvement in these areas.
- **Certain companies** receive far more complaints than others, suggesting a need for better customer service or clearer policies.
- **Most complaints are submitted online**, highlighting the importance of a user-friendly digital reporting process.
- **While many companies respond on time**, some still fail to do so, which affects consumer trust.
- **Complaint resolution varies**, with many closed without any relief to the consumer.

The dashboard makes it easy to explore this data visually and interactively. It can help companies, regulators, and consumers better understand complaint trends and take steps to improve services, policies, and consumer protection.

6. Future Scope

This project provides a strong starting point for understanding consumer complaints, but there are many ways it can be expanded and improved in the future:

- **Live Data Updates:** Connect the dashboard to real-time data sources from data.gov or the CFPB to keep the insights up to date.
- **Text Analysis:** Use natural language processing (NLP) to analyze the actual complaint texts for deeper insights into customer concerns and emotions.
- **Company Comparison:** Add advanced filtering to compare how different companies handle complaints over time.
- **Geographical Insights:** Include maps to show complaint patterns across states or regions.
- **Customer Satisfaction Tracking:** Add features to track how well companies resolve issues and how satisfied consumers are after filing complaints.
- **Mobile-Friendly Dashboard:** Create a version of the dashboard that works smoothly on phones and tablets for easier access by all users.

These improvements would make the project more powerful and helpful for regulators, companies, and consumers looking to improve financial services.

7.) Links :-

LinkedIn Post Link :

<https://www.linkedin.com/feed/update/urn:li:activity:7317244735205617664/>

The screenshot displays a LinkedIn post by Shezal Fatima, a B.Tech-CSE student at Lovely Professional University. The post features a dashboard with various charts and graphs, titled "Here is my newly updated dashboard.....". The post has received 597 impressions and 4 reactions (2 likes, 1 comment, and 1 repost). The author's profile is visible on the left, showing her name, education, and a "Try now for ₹0" button. The right sidebar contains a promoted post for aramco and a footer with LinkedIn navigation links and copyright information.

Profile: Shezal Fatima, B.Tech-CSE | Python | C | C++ | Java | Phagwara, Punjab | Lovely Professional University

Post Content: Here is my newly updated dashboard.....

Engagement: 597 impressions, 4 reactions (2 likes, 1 comment, 1 repost)

Comments:

- Gautam Kumar • 1st: Great work
- Krity kumari • 1st: Great work

Footer: LinkedIn Corporation © 2025